

KIMHYONGGWON, Korea, Democratic People's Republic ofWMO: **470220**Lat: **40.817N**Lon: **128.150E**Elev: **1200**StdP: **87.72**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.7	-23.8	-28.9	0.3	-24.7	-27.2	0.4	-23.0	12.2	-13.6	9.3	-18.1	1.3	320	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	26.6	19.5	25.1	18.9	23.7	18.1	21.4	25.1	20.3	23.8	19.4	22.5	1.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.2	17.2	23.9	19.2	16.1	22.6	18.3	15.2	21.5	68.7	25.4	64.5	23.8	61.0	22.6	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.3	6.0	-28.8	29.5	2.2	1.7	-30.4	30.8	-31.7	31.8	-32.9	32.7	-34.5	34.0		
(5)			-28.6	22.8	2.2	1.5	-30.2	23.9	-31.6	24.8	-32.8	25.6	-34.4	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-14.1	-11.0	-4.2	3.7	10.5	14.7	18.4	17.8	12.1	5.5	-3.6	-11.7
	DBStd	12.02	4.36	4.77	5.18	4.52	3.63	2.51	2.55	2.79	3.05	3.59	5.24	4.74
	HDD10.0	3244	747	587	441	192	37	1	0	0	14	144	409	673
	HDD18.3	5564	1006	821	699	438	243	109	31	43	187	398	659	931
	CDD10.0	786	0	0	0	4	53	144	259	242	78	4	0	0
	CDD18.3	61	0	0	0	0	1	2	32	26	0	0	0	0
	CDH23.3	393	0	0	0	2	31	50	188	120	3	0	0	0
CDH26.7	49	0	0	0	0	4	3	28	14	0	0	0	0	
Wind	WSAvg	1.6	1.8	1.9	2.1	2.1	2.1	1.3	1.0	1.0	1.2	1.4	1.7	1.7
Precipitation	PrecAvg	679	7	11	15	32	69	104	166	139	73	29	23	12
	PrecMax	1108	34	61	35	68	337	167	340	293	204	69	73	47
	PrecMin	496	1	0	2	8	17	45	72	25	9	0	1	1
	PrecStd	166	6	13	8	15	59	38	73	64	51	18	19	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.1	5.9	13.6	21.7	26.3	26.9	29.6	28.5	23.5	19.7	12.1	3.6
		MCWB	-3.2	2.5	6.5	10.2	15.5	16.9	21.4	22.3	16.3	11.2	6.5	0.5
	2%	DB	-2.6	2.7	9.7	17.7	23.6	24.8	26.9	26.1	21.6	17.1	9.3	0.7
		MCWB	-5.0	-0.7	3.9	8.8	13.6	16.4	20.6	20.4	15.3	9.6	4.5	-1.7
	5%	DB	-4.5	0.1	6.9	15.0	21.2	22.9	25.3	24.5	20.1	14.9	6.6	-1.7
		MCWB	-6.5	-2.7	2.0	7.6	12.1	15.6	20.0	19.9	14.4	8.4	3.1	-3.6
	10%	DB	-6.4	-2.2	4.4	12.6	18.9	21.2	23.8	23.0	18.5	12.6	4.3	-3.8
		MCWB	-8.0	-4.5	0.6	6.0	11.3	15.0	19.0	19.1	13.6	7.4	1.6	-5.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.0	2.5	7.2	12.1	16.8	19.4	22.9	23.3	18.4	12.5	7.2	1.1
		MCDB	-0.6	4.5	12.7	18.2	23.7	24.1	27.5	27.6	21.3	17.8	10.3	3.1
	2%	WB	-4.9	-0.5	4.5	9.6	14.7	17.9	21.6	21.5	16.9	10.8	5.3	-1.6
		MCDB	-2.9	2.3	8.8	15.9	21.8	22.2	25.6	24.8	20.1	15.2	8.6	0.4
	5%	WB	-6.3	-2.7	2.5	8.0	13.3	16.8	20.7	20.4	15.6	9.3	3.5	-3.5
		MCDB	-4.7	0.0	6.5	13.8	19.5	21.1	24.4	23.5	18.6	13.7	6.3	-1.7
	10%	WB	-8.0	-4.5	0.6	6.5	12.1	15.8	19.7	19.5	14.4	8.0	1.8	-5.4
		MCDB	-6.4	-2.2	4.1	11.9	17.7	20.1	23.0	22.5	17.3	11.8	4.2	-3.9
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.0	11.9	12.4	13.0	11.6	9.5	9.4	12.0	13.4	11.9	12.2
		MCDBR	14.6	15.6	15.0	17.1	18.0	15.4	12.3	11.4	14.2	16.9	14.9	14.2
		MCWBR	12.5	12.3	10.2	10.1	9.4	8.0	7.0	6.8	8.3	10.0	10.6	11.9
	5% WB	MCDBR	14.4	14.9	14.5	16.0	15.8	12.9	10.6	10.0	11.2	15.0	13.8	13.8
		MCWBR	12.5	12.1	10.4	10.3	9.3	7.8	6.6	6.6	7.5	9.6	10.2	11.7
Clear-Sky Solar Irradiance	taub		0.311	0.367	0.440	0.507	0.536	0.564	0.507	0.461	0.408	0.401	0.351	0.311
	taud		2.140	1.998	1.864	1.785	1.780	1.777	2.008	2.130	2.206	2.099	2.145	2.173
	Ebn at Noon		821	817	794	770	759	738	778	804	824	772	763	786
	Edh at Noon		104	141	182	211	217	219	172	148	128	127	104	93
All-Sky Solar Radiation	RadAvg		2.48	3.44	4.50	5.06	5.40	5.05	4.58	4.52	4.24	3.42	2.38	2.10
	RadStd		0.13	0.19	0.27	0.34	0.49	0.47	0.53	0.39	0.41	0.27	0.30	0.13